Task 1: Create an alert rule for CPU usage above 70%

1. Open AWS CloudWatch.
2. Select Alarms → Create alarm.
3. Choose the EC2 CPUUtilization metric for your instance.
4. Set the threshold to greater than 70%.
5. Configure a five-minute evaluation period.
6. Create the alarm named CPU-Usage-70-Percent.

Task 2: Configure email notifications

1. Create or select an SNS topic.
2. Subscribe your email address to the topic.
3. Confirm the subscription through the email confirmation message.
4. Attach the topic to the CPU alarm.
5. Verify the alarm action is enabled.

Task 3: Simulate high CPU usage

On a Linux test instance, use:

- sudo apt update
- sudo apt install stress -y
- stress --cpu 2 --timeout 360

After running the test:
1. Open the CloudWatch CPU utilization graph.
2. Wait for the next metric update.
3. Review the alarm state.
4. Check your email for the notification.
5. Capture a screenshot of the metric graph and alarm state.

Task 4: Add a memory usage condition

1. Install and configure the CloudWatch Agent on your Linux EC2 instance.
2. Enable memory metrics, such as mem_used_percent, and verify that they appear in CloudWatch.
3. Create a memory alarm using the published metric.
4. Set the threshold to greater than 80%.
5. Configure the same SNS notification topic.
6. Test the CPU and memory alarms separately using an appropriate test workload.

Task 5: Three improvements to the alerting system

1. Reduce false alarms

Require a threshold to be exceeded across multiple evaluation periods rather than triggering on a single temporary spike. This reduces unnecessary notifications caused by brief workload increases.

2. Prioritize notifications

Use different alert levels for warning and critical conditions. Send urgent alerts to the responsible team and include the resource name, metric, threshold, time and a link to the monitoring dashboard.

3. Monitor trends and recovery

Track CPU, memory, storage, availability and error rates together. Configure notifications for recovery and review historical metrics to identify recurring problems before they affect customers.